Advanced Empirical Finance: Topics and Data Science

Multifactor models and portfolio sorts

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Today

- Assignment 1 discussion (first hour)
- Sorting stocks on beta
- From one factor to many

Assignment 1 discussion

Sorting stocks on beta

Where the CAPM stands

- **Lecture 6:** ten industries, 1960–2025. Higher betas came with barely higher average returns, but the GRS test could not reject the CAPM: the industries are too similar to each other and to the market
- **Today:** all US stocks in CRSP, sorted into 10 portfolios by their beta
- The betas are those of exercise 4: each month, a CAPM regression on the last 60 months (at least 48)

> The CAPM says: the higher the beta, the higher the average return, one for one with the market premium

Betas for every stock and month

R

```r
betas <- crsp_monthly |>
  inner_join(factors_ff3_monthly, join_by(date)) |>
  estimate_betas("ret_excess ~ mkt_excess", months(window_length), min_obs = min_months)
```

- `estimate_betas()` from tidyfinance runs the regression on every 60-month window of every stock at once; exercise 4 does it window by window, with the same numbers
- 2,401,783 betas for 18,498 stocks, 1963–2025; the median beta is 1.04

Betas for every stock and month

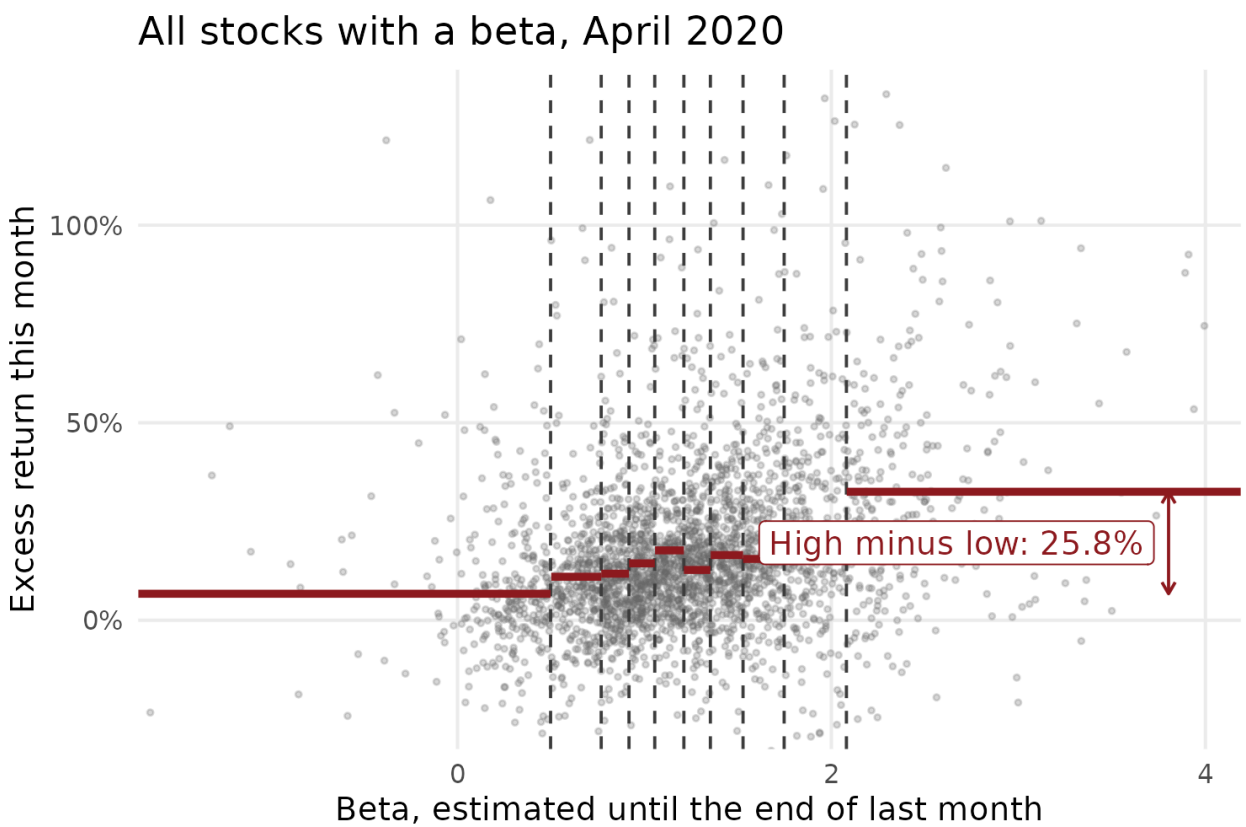
Python

```
betas = tf.estimate_betas(
    crsp_monthly.join(factors_ff3_monthly, on="date"),
    "ret_excess ~ mkt_excess", lookback=f"{window_length}mo", min_obs=min_months,
)
```

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What a portfolio sort does

- Each month, rank the stocks by **last month's beta**, which investors know when they form the portfolios
- **Breakpoints** split them into 10 groups of equal size
- Each portfolio's return is the **value-weighted** average return of its stocks
- **High minus low:** the top portfolio minus the bottom portfolio



In April 2020 the market rose by 13.6%: high-beta stocks rose most. Repeat every month and average

Sorting on beta in code

R

```r
# Sort on last month's beta, which investors know when they form portfolios
beta_lag <- betas |>
  transmute(permno, date = date %m+% months(1), beta_lag = beta_mkt_excess)

beta_portfolios <- crsp_monthly |>
  inner_join(beta_lag, join_by(permno, date)) |>
  compute_portfolio_returns(
    "beta_lag", "univariate",
    breakpoint_options_main = breakpoint_options(n_portfolios = n_portfolios)
  ) |>
  left_join(factors_ff3_monthly, join_by(date))
```

`compute_portfolio_returns()` sets the breakpoints each month and returns the value-weighted (`ret_excess_vw`), equal-weighted and capped value-weighted excess returns of each portfolio

Sorting on beta in code

Python

```python
# Sort on last month's beta, which investors know when they form portfolios
beta_lag = betas.select(
    "permno", pl.col("date").dt.offset_by("1mo"), beta_lag=pl.col("beta_mkt_excess")
)

beta_portfolios = (
    tf.compute_portfolio_returns(
        crsp_monthly.join(beta_lag, on=["permno", "date"]),
        "beta_lag", "univariate",
        breakpoint_options_main=tf.breakpoint_options(n_portfolios=n_portfolios),
    )
    .join(factors_ff3_monthly, on="date", how="left")
)
```

`compute_portfolio_returns()` sets the breakpoints each month and returns the value-weighted (`ret_excess_vw`), equal-weighted and capped value-weighted excess returns of each portfolio

The CAPM for each portfolio

R

```
capm <- beta_portfolios |>
  nest(data = -portfolio) |>
  mutate(
    fit = map(data, \(d) {
      estimate_model(d, "ret_excess_vw ~ mkt_excess", output = c("coefficients", "tstats"))
    }),
    alpha = map_dbl(fit, \(f) f$coefficients$intercept),
    t_alpha = map_dbl(fit, \(f) f$tstats$intercept),
    beta = map_dbl(fit, \(f) f$coefficients$mkt_excess),
    mean = map_dbl(data, \(d) mean(d$ret_excess_vw))
  ) |>
  select(portfolio, alpha, t_alpha, beta, mean) |>
  arrange(portfolio)
```

The same regression as for the industries in lecture 6, now for the 10 beta portfolios, with `estimate_model()` from tidyfinance

The CAPM for each portfolio

Python

```
rows = []
for (portfolio,), data in beta_portfolios.group_by("portfolio"):
    fit = tf.estimate_model(data, "ret_excess_vw ~ mkt_excess", output=["coefficients", "tstats"])
    rows.append({"portfolio": portfolio, "alpha": fit["coefficients"]["intercept"][0],
                 "t_alpha": fit["tstats"]["intercept"][0],
                 "beta": fit["coefficients"]["mkt_excess"][0],
                 "mean": data["ret_excess_vw"].mean()})
capm = pl.DataFrame(rows).sort("portfolio")
```

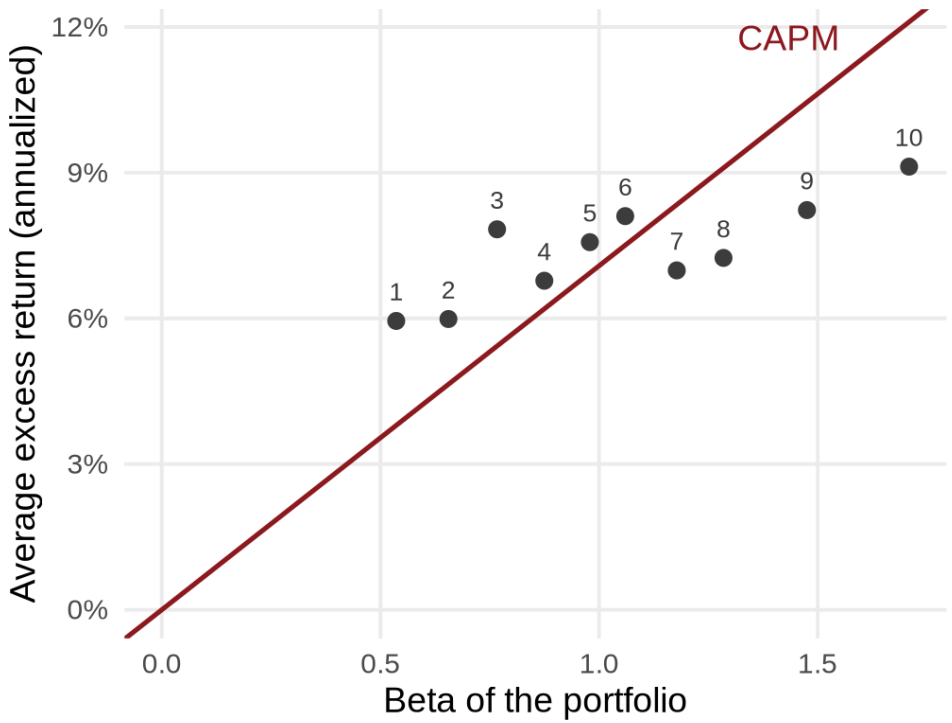
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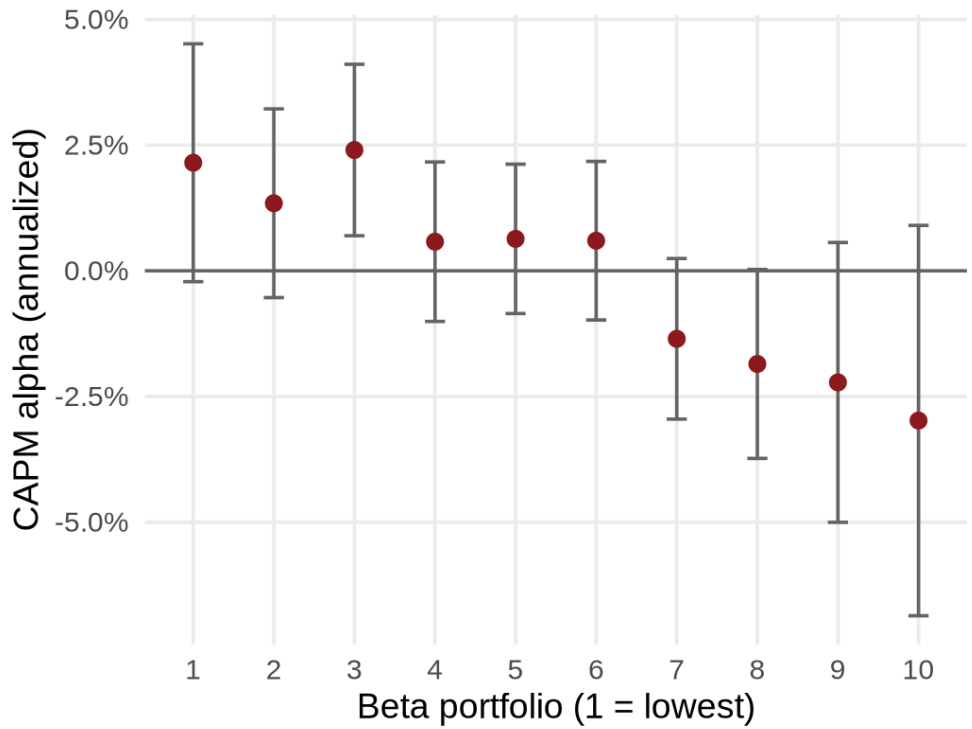
Which stocks have the higher alpha?

Ten portfolios sorted on beta, 1964–2025: which have the higher CAPM alpha?

- High-beta stocks
- Neither
- Low-beta stocks

Ten beta portfolios





- Betas from 0.54 to 1.71, average excess returns only from 5.9% to 9.1% a year. The CAPM predicts 3.8% to 12.1%: the line is too flat
- High minus low: alpha -5.1% a year (t = -1.87). GRS = 1.29, p-value 0.23: a clear pattern, but 1964–2025 is not enough to reject the CAPM with ten correlated portfolios

From one factor to many

Whiteboard: many factors

Suppose the SDF is linear in K factors, $m = a - b'f$. What does $E[m\, R_i^e] = 0$ imply for expected returns?

$$E[R_i^e] = \beta_i' \lambda, \qquad \beta_i = \mathrm{Var}(f)^{-1}\mathrm{Cov}(f, R_i^e), \qquad \lambda = \frac{\mathrm{Var}(f)\, b}{E[m]}$$

- If the factors are excess returns, each factor prices itself: $\lambda = E[f]$. In $R_{i,t}^e = \alpha_i + \beta_i' f_t + \varepsilon_{i,t}$, the model says $\alpha_i = 0$ for every asset, and GRS tests all alphas at once
- **Arbitrage pricing theory** (Ross, 1976): if returns have a factor structure and there is no arbitrage, expected returns are approximately $\beta_i' \lambda$

Where do factors come from?

- We do not observe betas. A characteristic z_i (size, book-to-market, past returns) is useful when it tells us something about the betas: $E[R_i^e] = \beta(z_i)'\lambda$
- Value firms are often close to distress and lose most in bad states: high beta, high premium
- A **portfolio sort turns a characteristic into a factor**: the top portfolio minus the bottom portfolio. Fama & French (1993) add SMB (small minus big) and HML (high minus low book-to-market) to the market: the three-factor model
- Today's sort on beta itself: low-beta stocks earn too much, high-beta stocks too little. Frazzini & Pedersen (2014) find the same in many markets and asset classes and turn it into a factor, *betting against beta*

Anomalies

Anomaly	Finding	Key reference(s)
Betting against beta	α declining in β across many asset classes	Frazzini & Pedersen (2014)
Size	Small stocks outperform large stocks	Banz (1981); Fama & French (1992)
Value	High book-to-market stocks outperform growth stocks	Rosenberg, Reid & Lanstein (1985); Fama & French (1992)
Momentum	Intermediate-term winners outperform losers	Jegadeesh & Titman (1993); Carhart (1997)
Liquidity	Illiquid stocks earn a return premium	Pástor & Stambaugh (2003)

- Hundreds more have been published since: the *factor zoo*. Which of them survive is a question for the lecture on reproducibility; how to use many characteristics at once, for the machine learning lectures